

MANCHESTER-BOSTON, NH, USA

WMO: 743945

Lat: 42.930N

Lon: 71.436W

Elev: 67

StdP: 100.52

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14710

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-13.9	-25.0	0.4	-14.2	-22.4	0.5	-11.2	10.3	-6.8	9.2	-5.4	3.6	330	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	32.8	22.0	31.3	21.3	29.8	20.6	24.0	29.4	23.1	28.1	22.3	27.2	4.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.2	26.4	21.6	16.4	25.7	20.9	15.6	25.1	71.9	29.4	68.7	28.2	65.8	26.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	8.0	7.2	DB	-20.3	35.7	3.0	1.7	-22.5	36.9	-24.2	38.0	-25.9	38.9	-28.1	40.2
			WB	-20.7	25.6	2.7	1.2	-22.7	26.5	-24.3	27.3	-25.9	28.0	-27.8	28.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-3.7	-2.1	2.3	8.9	14.9	20.0	23.2	22.2	18.0	11.3	5.2	-0.3	(6)
	DBStd	10.32	5.87	5.33	5.15	4.63	4.31	3.98	3.04	2.94	3.86	4.20	4.73	4.90	(7)
	HDD10.0	1600	424	341	245	73	6	0	0	0	1	34	157	321	(8)
	HDD18.3	3439	682	573	498	284	125	27	2	4	52	222	393	577	(9)
	CDD10.0	1623	1	1	6	41	158	301	409	378	241	74	13	1	(10)
	CDD18.3	423	0	0	1	2	19	78	153	124	43	3	0	0	(11)
	CDH23.3	3816	0	0	7	47	256	728	1393	1022	332	29	1	0	(12)
	CDH26.7	1313	0	0	2	13	85	258	526	331	95	3	0	0	(13)
(14) Wind	WSAvg	3.0	3.1	3.3	3.5	3.5	3.1	2.8	2.6	2.5	2.5	2.7	2.9	3.0	(14)
(15) Precipitation	PrecAvg	1116	81	76	97	92	92	97	86	93	94	107	97	103	(15)
	PrecMax	1620	154	262	282	234	329	275	206	227	220	354	154	166	(16)
	PrecMin	768	31	33	19	19	19	24	33	8	34	12	36	30	(17)
	PrecStd	211	30	43	55	53	60	64	46	54	51	74	31	31	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	15.7	22.7	28.5	32.2	34.1	34.9	34.0	32.3	26.5	21.1	16.3	(19)
		MCWB	13.5	10.6	14.0	15.6	19.1	22.3	23.6	23.1	21.6	17.9	16.2	11.6	(20)
	2%	DB	10.2	11.1	17.0	23.7	29.1	32.1	32.8	32.0	29.4	23.4	17.8	12.1	(21)
		MCWB	7.1	6.6	10.5	13.8	17.9	21.6	22.5	21.9	20.6	17.0	13.8	9.3	(22)
	5%	DB	6.8	7.8	12.8	20.3	26.2	29.6	31.3	30.1	27.1	21.1	15.5	9.1	(23)
		MCWB	4.2	4.1	6.9	12.2	16.3	20.2	21.7	21.0	19.6	16.0	11.6	6.3	(24)
	10%	DB	3.9	5.2	9.9	17.3	23.3	27.6	29.7	28.4	24.8	18.4	12.7	6.8	(25)
		MCWB	1.5	2.1	5.3	10.1	15.0	19.0	21.0	20.4	18.5	13.9	9.4	4.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	12.4	15.1	18.1	21.3	24.5	25.3	24.8	23.6	20.6	17.5	13.4	(27)
		MCDB	15.0	14.8	21.3	23.7	29.2	31.6	32.4	31.0	28.0	23.3	19.4	15.4	(28)
	2%	WB	7.8	7.3	11.5	15.5	19.8	23.0	24.1	23.6	22.3	18.5	14.9	9.8	(29)
		MCDB	9.2	9.7	15.4	21.7	25.7	28.8	29.9	28.4	26.6	21.9	17.3	11.4	(30)
	5%	WB	4.2	4.3	8.0	13.3	18.1	22.0	23.2	22.7	21.2	16.6	12.2	6.7	(31)
		MCDB	6.5	7.3	11.5	18.6	23.9	27.4	28.5	27.3	25.0	20.1	14.2	8.6	(32)
	10%	WB	1.9	2.5	5.6	10.8	16.5	20.7	22.4	21.9	20.0	14.8	9.9	4.2	(33)
		MCDB	3.6	4.9	9.3	16.1	22.0	25.1	27.4	26.5	23.7	17.9	12.6	6.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	9.5	11.0	11.3	10.7	10.6	10.5	10.6	10.1	8.7	7.9	(35)
		MCDBR	11.3	12.3	14.1	16.5	16.0	13.7	12.6	12.4	13.0	14.0	12.1	11.3	(36)
	5% WB	MCWBR	9.5	9.0	8.9	8.6	7.3	5.7	4.6	4.8	6.0	8.0	8.8	9.2	(37)
		MCWBR	10.8	11.6	13.2	14.5	13.7	11.7	10.8	10.3	10.4	11.7	10.4	10.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.284	0.295	0.321	0.360	0.396	0.437	0.459	0.426	0.372	0.332	0.308	0.288	(40)
		taud	2.452	2.431	2.451	2.395	2.332	2.241	2.192	2.267	2.401	2.523	2.548	2.533	(41)
	Ebn at Noon		863	911	925	907	878	839	815	831	857	857	830	830	(42)
		Edh at Noon	73	89	99	113	124	136	142	128	103	81	68	63	(43)
(44) All-Sky Solar Radiation	RadAvg		1.75	2.61	3.65	4.60	5.08	5.49	5.79	5.23	4.23	2.73	1.87	1.41	(44)
	RadStd		0.11	0.20	0.24	0.50	0.61	0.52	0.37	0.27	0.28	0.26	0.17	0.15	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+36

Nomenclature: See separate page